



Research Design & Validation

Rendani Mamphiswana, PhD
Innovation Director at iCatalysis Hub

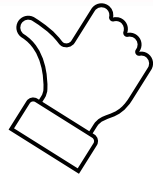
27 August 2026

SATNAC 2026 Industry Solutions Challenge



Innovation Enablement & Solution Development: Masterclass Series
Timeline

20 Aug	27 Aug	3 Sept	10 Sept	17 Sept	24 Sept
SESSION 01 Problem Framing & Concept Design	SESSION 02 Research Design & Validation	SESSION 03 Business Modelling & Value Design	SESSION 04 Technology & Architecture Design	SESSION 05 Prototyping & MVP Development	SESSION 06 Executive Pitch Readiness



Every idea contains untested claims



Your team is usually claiming that:

1. A real and important problem exists
2. A defined group experiences it
3. Your intervention changes an outcome
4. Users and stakeholders will adopt it
5. Delivery is feasible in the intended context

What you will be able to do/ learning outcomes



Convert a solution concept into testable assumptions:

1. Match each key question to credible evidence and a feasible method
2. Define validation criteria before collecting data
3. Select success measures that inform a decision
4. Produce a concise research and validation plan



Research and validation

Research asks

- What is happening, for whom, where and why?

Validation asks

- Is the evidence strong enough to keep, change or stop an assumption?

The cycle

- Assumption → Question → Evidence → Decision → Next Test

Research creates evidence while validation informs decisions



Test the riskiest assumptions first

Desirability

- Do users need it, understand it and choose it?

Feasibility

- Can it work technically and operationally in the target setting?

Viability & Impact

- Can it sustain resources, create value and avoid unacceptable harm?

Put a decision inside the research question



Weak

- “Do students like our platform?”

Stronger

- “Among off-campus students with intermittent connectivity, does a low-data prototype enable completion of the priority task within five minutes, without assistance”.

Specify: User + context + behaviour/outcome + comparison or threshold

Match the method to the uncertainty



Desk research → Market, policy, technology and prior evidence.

Interviews → Needs, motivation, workflow and constraints.

Survey → Prevalence, distribution and patterns in a defined population.

Observation/usage data → Actual behaviour and process friction

Prototype/experiment → Causal mechanism, usability and performance.

Triangulate when one source cannot answer the decision

Sample for the context; protect the participant



Who must be included?

- Users, non-users, implementers, decision-makers and affected groups.

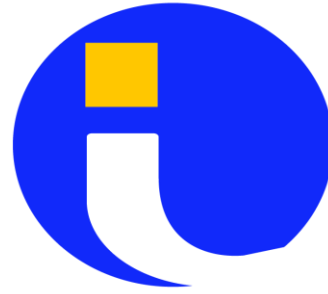
What variation matters?

- Country, language, institution, location, gender, disability, device and connectivity.

Ethical minimum

- Voluntary consent; data minimisation; privacy; secure storage; no avoidable harm

Define the decision rule before the test



For each critical assumption, record:

1. Indicator – What will be observed or measured?
2. Threshold – What results counts as sufficient?
3. Evidence quality – Whose data, how is it collected, under what conditions?
4. Decision – Pass, revise, retest or stop.

Predefined criteria reduces confirmation bias.

Measure learning now – and outcomes later



Leading measures – Early signals

- Task completion •Error rate •Time-on-task •Activation •Repeated use
- Willingness to pilot

Lagging measures – Value and impact

- Cost reduction •Service reliability •Retention access •Learning outcomes
- Emissions avoided

Guardrails – What must not worsen

- Privacy •Exclusion •Workload •Security •Affordability

Build a minimum viable validation plan



Assumption: What must be true?

Question: What do we need to know?

Method: What is the smallest credible test?

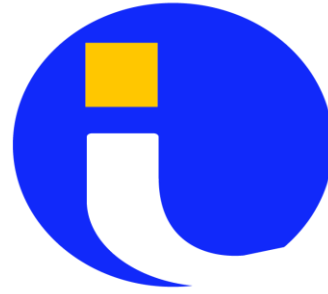
Sample: Who/what/where/how many?

Measure: What counts as sufficient?

Criterion: What counts as sufficient?

Decision: What will we do for each result?

Owner and date: Who completes the test, and when?



Worked case: Low-data support

Claim: The tool helps off-campus students to recover from interruptions and complete a priority task.

Assumption: The main barrier is workflow failure-not device access or data cost.

Evidence: 8-12 interviews + 6-8 observed prototype attempts

Measures: Completion, errors and workarounds

Decision: Proceed if the barrier is addressed and most users complete the task unaided.



Mixed evidence is useful evidence

If evidence supports the claim:

- Proceed cautiously, test under more realistic conditions.

If evidence is mixed:

- Segment users, inspect context, improve the test and revise the mechanism.

If evidence contradicts the claim:

- Change the assumption, concept or target user/or stop.

Always report limitations, negative findings or target user or stop



Your milestone deliverable

Submit a concise research and validation pack containing:

1. The decision and 2-5 priority assumptions.
2. Research questions linked to each assumption.
3. Methods, sample, recruitment and ethical safeguards.
4. Indicators, thresholds and success measures.
5. Evidence log: Source, finding, limitation and implication
6. Pass-revise-stop decisions, owners and dates

A credible plan is specific enough for another team to execute and arrive at a similar decision.



Thank you



+2710 003 7170



innovate@icatalysishub.com

22 Magwa Crecent, Gateway West,
1st Floor, Waterfall City, 1682